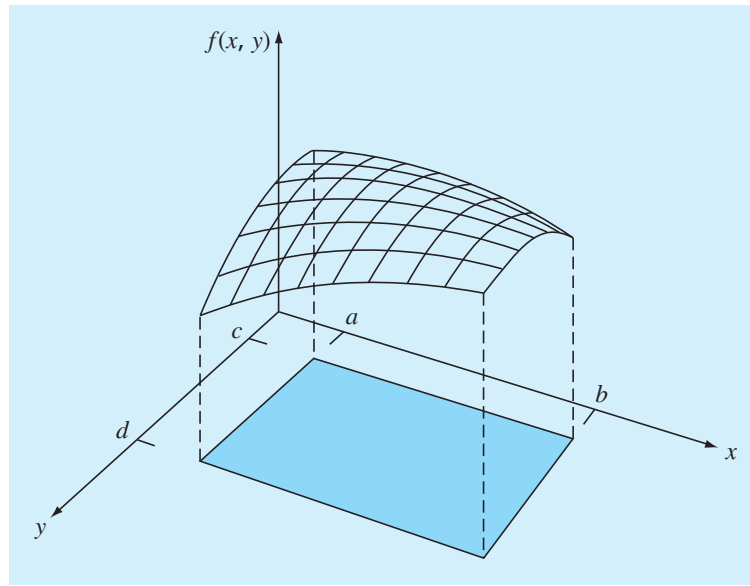


**FIGURE 19.16**

Double integral as the area under the function surface.

A numerical double integral would be based on the same idea. First, methods such as the composite trapezoidal or Simpson's rule would be applied in the first dimension with each value of the second dimension held constant. Then the method would be applied to integrate the second dimension. The approach is illustrated in the following example.

EXAMPLE 19.8 Using Double Integral to Determine Average Temperature

Problem Statement. Suppose that the temperature of a rectangular heated plate is described by the following function:

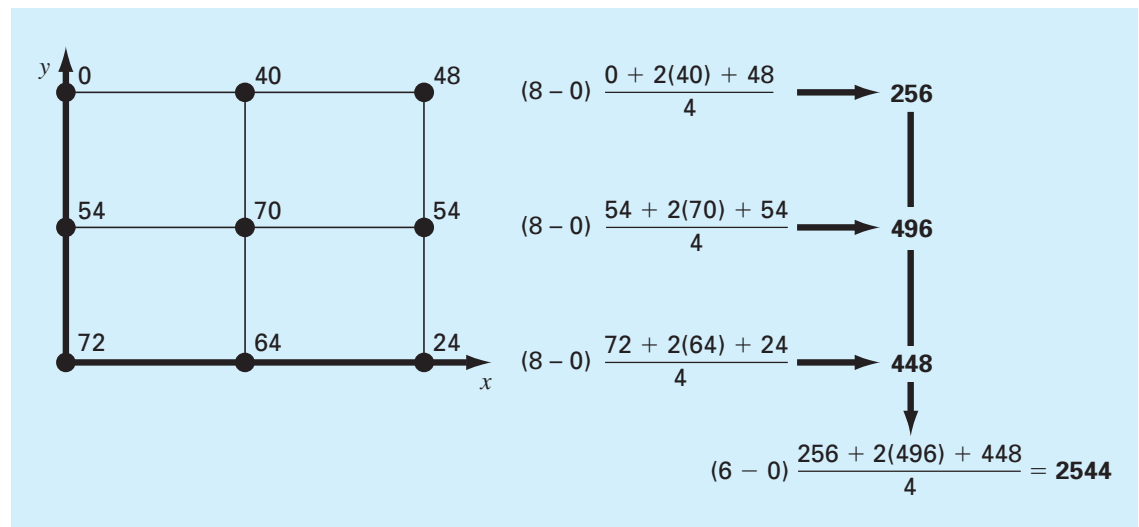
$$T(x, y) = 2xy + 2x - x^2 - 2y^2 + 72$$

If the plate is 8 m long (x dimension) and 6 m wide (y dimension), compute the average temperature.

Solution. First, let us merely use two-segment applications of the trapezoidal rule in each dimension. The temperatures at the necessary x and y values are depicted in Fig. 19.17. Note that a simple average of these values is 47.33. The function can also be evaluated analytically to yield a result of 58.66667.

To make the same evaluation numerically, the trapezoidal rule is first implemented along the x dimension for each y value. These values are then integrated along the y dimension to give the final result of 2544. Dividing this by the area yields the average temperature as $2544/(6 \times 8) = 53$.

Now we can apply a single-segment Simpson's $1/3$ rule in the same fashion. This results in an integral of 2816 and an average of 58.66667, which is exact. Why does this occur? Recall that Simpson's $1/3$ rule yielded perfect results for cubic polynomials. Since the highest-order term in the function is second order, the same exact result occurs for the present case.

**FIGURE 19.17**

Numerical evaluation of a double integral using the two-segment trapezoidal rule.

For higher-order algebraic functions as well as transcendental functions, it would be necessary to use composite applications to attain accurate integral estimates. In addition, Chap. 20 introduces techniques that are more efficient than the Newton-Cotes formulas for evaluating integrals of given functions. These often provide a superior means to implement the numerical integrations for multiple integrals.

19.8.1 MATLAB Functions: `dblquad` and `triplequad`

MATLAB has functions to implement both double (`dblquad`) and triple (`triplequad`) integration. A simple representation of the syntax for `dblquad` is

```
q = dblquad(fun, xmin, xmax, ymin, ymax, tol)
```

where q is the double integral of the function fun over the ranges from $xmin$ to $xmax$ and $ymin$ to $ymax$. If tol is not specified, a default tolerance of 1×10^{-6} is used.

Here is an example of how this function can be used to compute the double integral evaluated in Example 19.7:

```
>> q = dblquad(@(x,y) 2*x*y+2*x-x.^2-2*y.^2+72, 0, 8, 0, 6)
```

```
q =  
2816
```